included PCs, servers, camera, etc - the smartphone had constraints of power, space, memory, and budget. So, it is safe to say that the constraints of the IoT are similar.

We can expect a high level of integration that will come in next few decades. Some other expectations are:

Cloudless architecture to overcome design constraints: Cloudless technology will enhance various domains of the IoT. It will provide greater choice and agility while reducing cost for applications that incorporate state-of-the-art hardware, software, and data access. You can think of a variety of cloudless architectures, like mesh networks that offer better latency and privacy. In fact, there are some companies already thinking about piloting peer-to-peer cloudless architectures. Therefore, privacy and reduced latency will be the biggest reason for the adoption of cloudless technology.

Self-organising network: A self-organising network (SON) uses artificial intelligence (AI), predictive analytics, and pre-optimised software algorithms to offer automated functions, such as self-configuration, self-optimisation, self-healing, and self-protection. SONs simplify network administration by enabling the creation of a plug-and-play environment for a network task. There are over fifty different wireless protocols that serve a variety of different needs. Bluetooth has a different role than Wi-Fi and so does the sub-gigahertz protocol that gives you a range of a few kilometres.

The benefit of implementing SON is that it will allow us to change the protocols on the fly. For example, the transmitter radios can automatically adjust parameters such as transmission power to avoid interference and maximise both coverage and capacity. It also provides better security as the self-organising network automatically defends itself from penetration by unauthorised users.

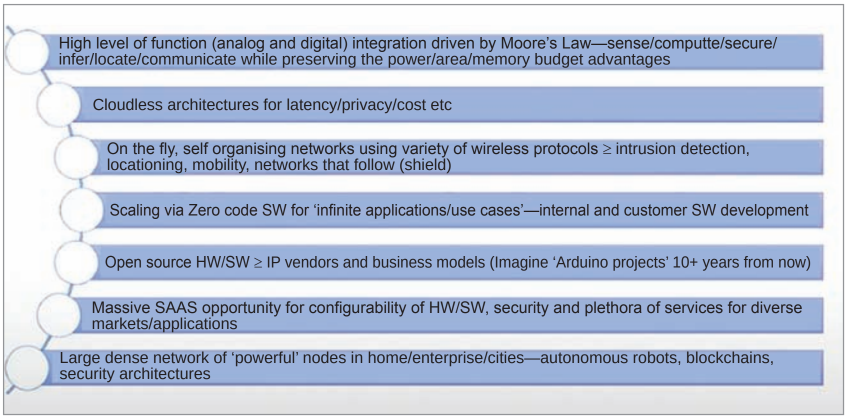

Figure 2: IoT future (Decade+) possibilities: Infinite and open to anyone's imagination

Open source software: The other very interesting thing that is bound to happen in software is scaling. In a world with infinite applications and use-cases, it is not possible to serve each individual customer's requirements. This is where a zero-code solution holds the potential to change the world. The zero-code solution will allow the customers to take the code, drag and drop connectors in a visual platform—and the code is outputted and the product is ready to go.

Growth with open source hardware: Open source hardware (OSH) is free-to-use physical artefacts of technology that can be made and used by anyone without any copyright infringement. There is a big open source community supported by electronics makers and DIYers all around the globe. One such example is of Arduino projects for microcontrollers and NodeMCU, which is an open source IoT platform. OSH is bound to grow and result in a variety of different business models that will disrupt the industries. Furthermore, it will empower individuals to create IoT solutions, thus democratising IoT.

Massive SAAS opportunity: The need for configurability of hardware and software creates a huge opportunity for software as a service (SaaS). Since there is a diverse market with specific requirements, IoT needs to meet a plethora of different demands, such as security and time relevance. So, there is an opportunity for a large dense network of powerful nodes. For example, every lighting pole present in the smart grid will have some sort of SoC that can self-organise and help you with location, sensing, etc. You can also think about running blockchain on it.

Although it seems far-fetched, we are talking about these in the future because we will have powerful computing power in these nodes. We are thinking about the 5G equivalent of IoT, which will have 8, 10, 12 cores offering significant compute power. Memory is only going to grow in all of these massive networks that would be available in the future.

These are all predictions, but what is certain is that the IoT holds the ability to disrupt a lot of industries and change the world for greater good. END

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